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Interventions for Students With Autism in Inclusive Settings: A Best-Evidence Synthesis and Meta-Analysis

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Students with autism spectrum disorder (ASD) are increasingly educated alongside typically developing peers in regular education environments. These students have impairments that may hinder their success in inclusive school settings and require individualized supports to improve outcomes. The purpose of this meta-analysis and best-evidence synthesis is to examine the characteristics of interventions for students with ASD in inclusive settings, offer quantitative analysis of intervention effects, examine potential moderating variables that influence outcomes, analyze the social validity of these interventions, and provide recommendations for practice and future research. The 28 included studies met the What Works Clearinghouse standards for group design and single-case design research. Studies focused mostly on social communication skills, produced moderate to large effects, and were generally found to be socially valid. Function-based interventions, visual supports, self-monitoring strategies, and peer-mediated interventions resulted in mostly large effects, and teacher delivered interventions produced the largest overall effects. More high-quality studies for students with ASD in inclusive school settings are needed to advance evidence-based practice for this population.

Public Significance Statement
This study analyzes interventions for students with autism spectrum disorder (ASD) in inclusive classroom settings. Function-based interventions, visual supports, self-monitoring, and peer-mediated interventions produced strong results. Interventions mostly targeted social communication skills and were considered feasible to implement in inclusive settings. Future studies that train teachers to implement these interventions, target additional skills, and include students with ASD with diverse characteristics are needed.

Keywords: autism spectrum disorder, intervention, inclusive education, meta-analysis

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Though prevalence rates vary across countries, the number of children diagnosed with autism spectrum disorder (ASD) continues to increase worldwide (Elsabbagh et al., 2012). In the United States, an estimated 1 in 59 school-age children is diagnosed with ASD (Baio et al., 2018), and federal mandates such as the Individuals with Disabilities Education Act (IDEA) require that students with ASD be educated alongside typically developing peers in regular education environments to the "maximum extent that is appropriate" (Individuals with Disabilities Education Improvement, 2014). International frameworks such as the Salamanca Statement and the World Declaration on Education for All also support inclusive education for students with disabilities and identify inclusion as a key aspect of educational reform (Ferguson, 2008; Peters, 2007; United Nations Educational, Scientific and Cultural Organization [UNESCO], 1994, 2002). Although many students with ASD still receive special education services in segregated classrooms specifically for students with disabilities, a growing proportion of these students receive special education services in inclusive settings where they are educated alongside their typically developing peers (Camargo et al., 2014; Koegel, Vernon, Koegel, Koegel, & Paullin, 2012; Watkins et al., 2015).

Students with ASD have impairments that may hinder their success in inclusive settings. Interaction and communication with classmates and teachers and participation in classroom activities can be difficult because of social communication deficits that are

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central to the diagnostic criteria of the disorder (American Psychiatric Association [APA], 2013). Restricted and repetitive interests or behaviors exhibited by students with ASD may also negatively affect both academic achievement and social relationships (APA, 2013; Lanovaz, Robertson, Soerono, & Watkins, 2013). In addition, challenging behaviors such as aggression or elopement that are common in students with ASD (Matson & Nebel-Schwalm, 2007) may be a barrier to successful inclusion in general education classrooms (Crosland & Dunlap, 2012; Dunlap, Iovannone, Wilson, Kincaid, & Strain, 2010; Emerson et al., 2001). In order for most students with ASD to experience academic, social, and behavioral gains in inclusive environments, appropriate supports must be in place, and teachers must implement scientific, evidence-based interventions for these students (Individuals with Disabilities Education Improvement, 2014). To promote generalization, these supports and interventions should occur in the student's natural environment, where the desired behaviors and skills are expected to occur (Crosland & Dunlap, 2012). Thus, it is vital that researchers identify which practices have met criteria to be considered evidence-based in inclusive settings and which of these strategies have been successfully implemented by practitioners.

Organizations such as the National Standards Project (NSP) and the National Professional Development Center on Autism Spectrum Disorders (NPDC) have identified 27 evidence-based practices for children and youth with ASD (see Wong et al., 2015), but these reviews did not focus specifically on inclusive educational settings and did not quantitatively analyze intervention results to determine for whom and under what conditions treatment would be most effective. Other reviews investigating interventions for students with ASD in inclusive settings have examined interventions targeting behaviors within a particular skill domain (e.g., social skills; Camargo et al., 2014), analyzed specific intervention strategies (e.g., peer-mediated interventions [PMI]; Watkins et al., 2015), or provided descriptive summaries of intervention strategies and trends in inclusive education (e.g., Crosland & Dunlap, 2012; Koegel et al., 2012). Together, these works have elucidated the diverse intervention approaches for students with ASD in inclusive classrooms, both at the student level (e.g., individualized behavioral interventions) and schoolwide level (e.g., schoolwide positive behavior support and response to intervention initiatives). However, to date there is no comprehensive examination of high-quality research studies for this population that quantitatively analyzes the efficacy of focused interventions for students with ASD in inclusive settings across skill domains. Furthermore, scant attention has been focused on the social validity of interventions (i.e., the social importance and acceptability intervention goals, procedures, and outcomes; Callahan, Henson, & Cowan, 2008), which is an important variable in determining whether or not practitioners will implement interventions with fidelity (Kennedy, 2002; Lang & Page, 2011). Finally, an examination of the variables that may moderate the effectiveness of these interventions (e.g., who delivers the intervention, the skill that is targeted) is necessary to determine for whom and under what conditions these practices will be most successful (Maggin, Chafouleas, Goddard, & Johnson, 2011; Wolery, Busick, Reichow, & Barton, 2010). These analyses are necessary to guide both future research and clinical practice aimed at improving outcomes for students with ASD.

Therefore, the purpose of this meta-analysis is to identify intervention studies for students with ASD in inclusive classroom settings that meet minimum standards for methodological rigor. Specifically, we: (a) summarize the characteristics of the interventions, (b) report effect sizes and offer an analysis of intervention effects, (c) examine potential moderating variables that influence outcomes, (d) analyze the social validity of these interventions, and (e) provide recommendations for practice and future research.

Method

Protocol Registration and PRISMA Guidelines

The protocol for this meta-analysis was registered with the PROSPERO International prospective register of systematic reviews (Watkins & O'Reilly, 2016) and was conducted in accordance with PRISMA guidelines (Moher, Liberati, Tetzlaff, & Altman, 2009).

Systematic Search Procedures

A search was conducted using the PsycINFO, Education Resources Information Center (ERIC), and Medline databases using the terms autis* or ASD or Asperger* or pervasive developmental disorder*; intervention or treatment or program; and inclus* or general education. The search was restricted to English language peer-reviewed studies published between 1997 through 2017, which is consistent with the 20-year period research synthesis organizations have recommended when examining in the literature (e.g., NPDC; What Works Clearinghouse [WWC], 2014). To identify relevant studies potentially missed by electronic search, ancestry searches of included articles were conducted, and citations that appeared potentially relevant were considered for inclusion. In addition, searches of literature reviews and meta-analyses that were returned from keyword database searches were conducted to identify further studies to consider for inclusion. A total of 1,585 records, published between 1997 and 2017, resulted from this multistep search procedure. Of these records, 126 abstracts from intervention studies were identified for screening. Eighty-four full-texts were then accessed for eligibility, with a total of 28 articles meeting the criteria for inclusion in the meta-analysis. Figure 1 depicts the search and screening process.

Inclusion Criteria

Inclusion criteria were developed before the literature search. Studies included in this review conducted focused interventions in an inclusive school setting that targeted a skill-based or behavioral outcome of a student age 3–21 years old diagnosed with autism spectrum disorder, including Asperger syndrome, autistic disorder, and pervasive developmental disorder not otherwise specified (PDD-NOS). If studies incorporated participants with other diagnoses, only the disaggregated data for participants with ASD were considered (e.g., Hughes et al., 2011). Studies that did not disaggregate data for participants with ASD were excluded (e.g., Carter et al., 2016).

Focused interventions were defined as operationally defined practices designed to address specific learner outcomes (Odom, Collet-Klingenberg, Rogers, & Hatton, 2010). Comprehensive

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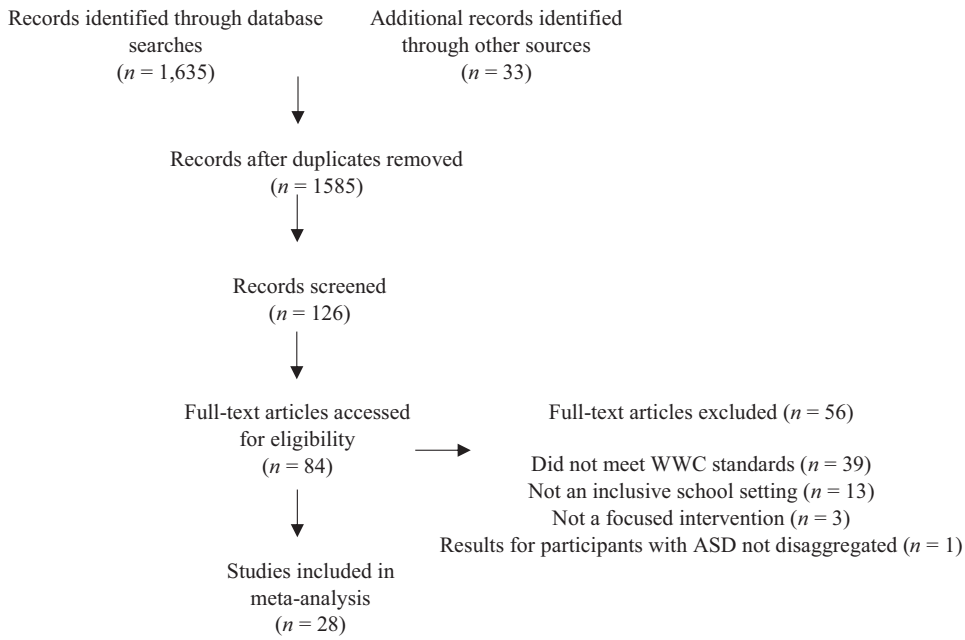


Figure 1. Flowchart of included studies.

treatment models (i.e., practices organized around a conceptual framework designed to broadly target learning or development; Wong et al., 2015) were excluded (e.g., Eldevik, Hastings, Jahr, & Hughes, 2012; Eikeseth, Klintwall, Jahr, & Karlsson, 2012; Strain & Bovey, 2011). Inclusive school settings were defined as those in which the student with ASD shared the context and activities with typically developing classmates (Watkins et al., 2015). As inclusion refers to the placement of special education students in general education settings (Camargo et al., 2014; Mesibov & Shea, 1996), studies that took place in a self-contained special education class were excluded (e.g., Banda & Hart, 2010; Kuhn, Bodkin, Devlin, & Doggett, 2008).

Finally, studies included in this meta-analysis must have used an experimental research design (i.e., group design or single-case design [SCD]) that allowed for direct analysis of the effect of the intervention on participant behavior or skills. Furthermore, the studies must have demonstrated sufficient methodological rigor according to the WWC standards for group design or SCD research (see WWC, 2014). The WWC provides a method for evaluating experimental research that has sufficient rigor and evidence of effectiveness, which can be used to identify studies to be included in research syntheses (Brock et al., 2017; King, Lemons, & Davidson, 2016; Maggin et al., 2011). Such best-evidence syntheses are necessary as they identify those interventions that are likely to result in positive outcomes for students with ASD, allowing practitioners to identify the most effective instructional strategies for these students and allowing researchers to contribute to and extend upon existing research (King et al., 2016; Slavin, 1995). Group design studies must have met minimum standards regarding randomization, sample attrition, and baseline equivalence. SCD studies must have met minimum standards for independent and dependent variables and the minimum standards of demonstration of effect over time.

Of the 84 full text articles screened for inclusion in the initial search process, 13 did not take place in an inclusive school setting (group design = 2; SCD = 11), 3 assessed comprehensive interventions rather than targeted interventions (group design = 3), and 1 did not report disaggregated results for participants with ASD (group design = 1); these studies were, thus, excluded. The first author and a coauthor then independently applied the WWC standards to the remaining 67 full text articles. This resulted in 94% agreement on studies meeting criteria, with discrepancies discussed between authors until 100% agreement was attained. A total of 28 studies (group design = 3; SCD = 25) met these criteria and are included in this meta-analysis.

Data Extraction

Each included study was summarized in terms of: (a) research design; (b) participant gender, age, ethnicity, and functioning level; (c) intervention agent, setting, intensity, type, and procedures; (d) skill domain and target behaviors; (e) intervention effectiveness, and (f) social validity indicators. Individual study summaries are provided in Table 1.

An estimation of participants' functioning level was determined by applying the schema provided by Reichow and Volkmar (2010). According to this schema, participants classified as lower functioning had limited or no verbal language skills and/or an IQ < 55. Participants classified at a moderate functioning level had basic verbal communication skills and/or an IQ of 55–85. Participants classified as high functioning were described as having high-functioning autism or Asperger's Syndrome, had well developed verbal communication and/or an IQ > 85.

The intervention strategies described by the authors of the studies were classified into interventions types (e.g., PMI, visual supports) to provide a gross estimation of the techniques and

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Table 1
Summary of Focused Interventions for Students With ASD in Inclusive Settings

Reference	Design	Participant characteristics	Setting and dosage	Agent, intervention type, and procedures	Skill domain and target behaviors	ES[a] and 95% CI
Blair, Umbreit, Dunlap, and Jung (2007)	MBD across activities	1 male; 6 years Low FXN Ethnicity NR	Classroom 10 min 3 × per week for 13 weeks, M = 13 sessions per activity	Teacher Function-based intervention: Antecedent and consequent modifications to instructional routines based on FBA data including choice-making, visual supports, PECS, and preferred interests; participant taught appropriate replacement skills through physical prompting, modeling, and time delay procedures	Challenging behavior: Out of seat, aggression, inappropriate vocalizations Classroom behavior: Engagement with materials, following directions Social communication skills: Communication with picture cue cards	Tau-U = 1 [.78, 1] NAP = 1 [.70, 1] Tau-U = .92 [.33, 1] NAP = 1 [.40, 1] Tau-U = 1 [.43, 1] NAP = 1 [.43, 1]
Bock (2007)	MBD across settings	1 male; 12 years High FXN Ethnicity NR	Classroom, cafeteria 5 × per week, M = 9 sessions per setting	Teacher Social script: Participant read three scripts related to different social-behavioral difficulties that detailed a "stop," "observe," "deliberate," and "act" (SODA) strategy to determine what to say and how to act in various social situations; teacher discussed SODA with student to check for comprehension	Social communication skills: Interaction Play skills: Game playing Classroom behavior: Cooperative learning	Tau-U = 1 [.52, 1] NAP = 1 [.52, 1] Tau-U = 1 [.58, 1] NAP = 1 [.52, 1] Tau-U = 1 [.40, 1] NAP = 1 [.4, 1]
Chan and O'Reilly (2008)	MBD across behaviors	2 males; 5 and 6 years High FXN Asian American and Caucasian	Classroom 10–20 min 1–4 × per week for 10 weeks, M = 10.5 sessions per behavior	Researcher Social Stories, role-play: Participants read Social Stories describing appropriate behaviors and answered comprehension questions; participant and researchers role-played skills presented in the Social Story; researcher used verbal prompts and praise during role-play sessions	Social communication skills: Interaction Challenging behavior: Inappropriate vocalizations and interactions Classroom behavior: Hand raising	Tau-U = .98 [.48, 1] NAP = .99 [.49, 1] Tau-U = .81 [.31, 1] NAP = .90 [.22, 1] Tau-U = 1 [.54, 1] NAP = 1 [.54, 1]
Cihak, Fahrenkrog, Ayres, and Smith (2010)	Reversal	3 males, 1 female; 6, 7, 8 years Low FXN Ethnicity NR	Classroom, hallways during transition periods 2–5 min, M = 13 sessions per participant	Teacher Video modeling: Participants trained to use a handheld device to access video self-modeling clips showing two-step tasks related to the morning routine; teachers verbally praised participants for correct video activation and completion of task; teachers used least-to-most prompting for incorrect responses	Classroom behavior: Transitions	Tau-U = .96 [.69, 1] NAP = .98 [.71, 1]
Cihak, Wright, and Ayres (2010)	Multiple probe across settings with embedded reversal	3 males; 11 and 13 years High FXN Ethnicity NR	Classroom 15 min, M = 21 sessions per participant	Teacher, paraprofessional Self-monitoring, visual cues: Participants trained to use a hand-held computer with static picture prompts to self-monitor behavior	Classroom behavior: Task engagement	Tau-U = 1 [.78, 1] NAP = 1 [.78, 1]
Clemons, Moore, Garrison-Kane, and Wills (2016)	Reversal	1 male; 17 years FXN NS White	Classroom 30 min, 10 sessions	Researcher Self-monitoring: Participant taught to use I-Connect self-monitoring app through modeling, praise, identification of examples of on-task behavior; participant used app in class and received reinforcer if criterion was met	Classroom behavior: On-task behavior	Tau-U = .80 [.33, 1] NAP = .99 [.50, 1]

(*table continues*)

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Table 1 (*continued*)

Reference	Design	Participant characteristics	Setting and dosage	Agent, intervention type, and procedures	Skill domain and target behaviors	ES[a] and 95% CI
Ganz and Flores (2008)	Changing criterion	3 males; 4 years High FXN Caucasian and Hispanic	Playroom 1–3 20 min sessions per day, 5 × per week, *M* = 10 sessions per participant	Researcher, peers PMI, script training: Participants taught to use scripted phrases for different play themes; peers trained to use visual instruction cards to initiate interaction with participants during play; peers modeled play behaviors for participants; adults used least to most prompting for participant use of scripted phrases	Social communication skills: Scripted and unscripted phrases, comments, responses, intervals with speech	Tau-U = .61 [.46, .76] NAP = .81 [.66, .96]
Gardner et al. (2014)	MBD across participants with embedded reversal	2 males; 14 and 18 years High, 1ow FXN Hispanic and European American	Classroom 30–40 min, 2–3 × per week, *M* = 7 sessions per participant	Teacher, peers Peer network: Peer networks supported by an adult facilitator were implemented in which peers and participants were given a target social skill goal and shared activity	Social communication skills: Interaction, social engagement	Tau-U = .81 [.53, 1] NAP = .94 [.66, 1]
Gena (2006)	MBD across participants	2 males, 2 females; 4 years Moderate FXN Ethnicity NR	Classroom; *M* = 10 sessions per participant	Teacher Prompting, reinforcement: Teacher used verbal and physical prompts for participants to engage in target behaviors; praise and social reinforcement for independent participant initiations and responses to peers; if participant exhibited challenging, he/she was removed from classroom and received compliance training from teacher	Social communication skills: Initiations, responses	Tau-U = .97 [.75, 1] NAP = .98 [.76, 1]
Haley, Heick, and Luiselli (2010)	Reversal	1 male; 8 years High FXN Caucasian	Classroom 40 min, 16 sessions	Teacher Function-based intervention, visual cues: Antecedent intervention developed based on FBA data; participant taught to use visual cues to discriminate when it was appropriate and inappropriate to engage in vocal stereotypy	RRB: Vocal stereotypy	Tau-U = 1 [.52, 1] NAP = 1 [.52, 1]
Hochman, Carter, Bottema-Beutel, Harvey, and Gustafson (2015)	MBD across participants	4 males; 15 and 17 years Moderate, low FXN Black, Latino, White	Cafeteria 30 min, 1 × per week, *M* = 7 sessions per participant	Teacher, peers Peer network: Peer networks supported by an adult facilitator were implemented in which peers and participants were given a target social skill goal and shared activity to participate in during lunch period; peers prompted interactions and modeled appropriate social skills	Social communication skills: Interaction, engagement	Tau-U = .76 [.58, .93] NAP = .89 [.71, 1]
Hughes et al. (2011)	Multiple probe across participants and settings	1 male, 2 females; 16, 20, and 21 years Moderate, low FXN Asian and White	Classroom, cafeteria 4–22 min pre training + 5 min, 5 × per week, *M* = 37 sessions per participant	Researcher, peers TIP, communication books, PMI: Trainer provided rationale for the social skill and used modeling, direct instruction, guided practice, and corrective feedback to teach participant to use communication book; peers trained to maintain interaction by initiating, expanding on topics, asking reciprocal questions, prompting participant use of communication book, and providing positive reinforcement	Social communication skills: Interaction, initiations	Tau-U = .97 [.87, 1] NAP = .98 [.88, 1]

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Table 1 (*continued*)

Reference	Design	Participant characteristics	Setting and dosage	Agent, intervention type, and procedures	Skill domain and target behaviors	ES[a] and 95% CI
Hughes et al. (2013)	Multiple probe across participants	2 males, 1 female; 16 and 17 years Moderate, low FXN Black and White	Classroom 90 min, 2–3 × per week, *M* = 26 sessions per participant	Researcher, peers PMI: Peers given suggestions for topics of conversation with participants based on shared interests; peers trained to set daily goals for number of initiations made to participants; peers trained to self-monitor and record number of interactions with participants	Social communication skills: Interaction	Tau-U = .91 [.59, 1] NAP = .95 [.64, 1]
Jung, Sainato, and Davis (2008)	MBD across participants	3 males; 5 and 6 years Moderate, low FXN Ethnicity NR	Classroom 10 min; 5 × per week up to 15 weeks, *M* = 35 sessions per participant	Researcher, peers PMI, high probability request sequences: Peers trained to interact with participants through role-play and discussion and taught to respond to participant's initiations; facilitator directed high probability requests to peer who modeled a response and then directed same request to a participant; facilitator directed a low probability request to a participant after 3 consecutive responses to a high probability request; praise provided for responding to requests	Social communication skills: Initiations, responses, interaction	Tau-U = .81 [.69, .93] NAP = .93 [.81, 1]
Kasari, Rotherman-Fuller, Locke, and Gulsrud (2012)	RCT	54 males, 6 females; 15 first graders, 18 second graders, 8 third graders, 11 fourth graders, 8 fifth graders High FXN 28 Caucasian, 13 Latino, 10 Asian, 6 Other, 3 African American	Playground CHILD and PEER intervention: 20 min, 2 × per week, 6 weeks total	Researcher, peers Social skills group (CHILD intervention): Students with ASD received didactic instruction with modeling and role-play of social skills from trained interventionist PMI (PEER intervention): Peers were taught strategies to engage children with ASD on the playground by using direct instruction, modeling, role play, and rehearsal	Social communication skills, challenging behavior: Social network salience, STRS closeness, conflict score, playground engagement, isolation during recess	*d* = .44 [.07, .94] *r* = .21 [.03, .42]
Kasari et al. (2016)	RCT	109 males, 28 females; 6–9 years Moderate, high FXN 62 Caucasian, 28 Asian, 20 Hispanic, 10 African American, 10 NS, 7 Other	Playground SKILLS and ENGAGE interventions: 30–45 min, 2 × per week, 8 weeks total	Researcher, peers Social skills group (SKILLS intervention): Didactic instruction in social skills with opportunities to practice skills, positive reinforcement for skill use, and homework assignments. Social skills group, PMI (ENGAGE intervention): Social skills group with children with ASD and peer models with shared interests; children participated in conversational exercises, structured games, free play, storytelling, and music	Social communication skills: Engagement with peers, joint engagement, isolation during recess, site effects (time spent in engagement)	*d* = .34 [.18, .50] *r* = .17 [.09, .24]
Kern and Aldridge (2006)	MBD across participants	4 males; 3 and 4 years Moderate, low FXN African American and European American	Playground 10 min, 5 × per week up to 34 weeks, *M* = 25 sessions per participant	Teacher, peers Music therapy, PMI: Individualized songs about turn-taking, choice-making, and body contact were taught to participants; teacher used verbal and physical prompts and praise to support play between participants and peers; peers trained to model play routines and praise participants	Social communication skills: Interaction	Tau-U = .94 [.78, 1] NAP = .97 [.80, 1]

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Table 1 (*continued*)

Reference	Design	Participant characteristics	Setting and dosage	Agent, intervention type, and procedures	Skill domain and target behaviors	ES[a] and 95% CI
Koegel, Vernon, Koegel, Koegel, and Paullin (2012)	MBD across participants with embedded reversal	3 males; 11, 13, and 14 years High FXN Ethnicity NR	Classroom, courtyard 2–3 × week, *M* = 13 sessions per participant	Researcher Peer network: Interests of participants incorporated into social club with peers led by an adult facilitator	Social communication skills: Initiations, engagement	Tau-U = .96 [.75, 1] NAP = .99 [.78, 1]
Koegel, Kim, Koegel, and Schwartzman (2013)	MBD across participants	6 males, 1 female; 14, 15, and 16 years high FXN Biracial, Caucasian, Hispanic	Classroom, school yard 30 min, 1 × per week for up to 10 weeks, *M* = 13 sessions per participant	Researcher Peer network: Preferred interests of participants incorporated into social clubs with peers led by an adult facilitator	Social communication skills: Initiations, engagement	Tau-U = .76 [.59, .93] NAP = .90 [.73, 1]
Kretzmann, Shih, and Kasari (2015)	RCT	18 males, 6 females; 6–11 years FXN NS 19 White, 4 Hispanic, 1 Black	Playground 16 para-professional training sessions; participant intervention sessions NS	Paraprofessionals Modeling, behavioral strategies NS (Remaking Recess intervention): Paraprofessionals trained to model interactions with peers and support engagement of students with ASD Waitlist control: Delayed intervention	Social communication skills: Peer engagement	*d* = 1.27 *r* = .54 Confidence intervals for the *d* and *r* values could not be calculated as no measure of variability was reported.
Loftin, Odom, and Lantz (2008)	MBD across participants	3 males; 9 and 10 years High FXN Ethnicity NR	Cafeteria 30 min, *M* = 10 sessions per participant	Researcher, peers PMI, initiation training, self-monitoring: Peers taught to reinforce initiations from participants in conversation or play; participants taught to initiate to peers using task analysis involving modeling, repeated trials, prompts, and reinforcement; participants taught to self-monitor initiations and provided tangible reinforcement contingent to correct response	Social communication skills: Initiations, interaction RRB: Motor stereotypy	Tau-U = .89 [.66, 1] NAP = .94 [.71, 1] Tau-U = .98 [.64, 1] NAP = .99 [.65, 1]
Mason et al. (2014)	MBD across participants	3 males; 6 and 8 years High, moderate FXN Ethnicity NR	Playground 10 min, 3 × per week, *M* = 13 sessions per participant	Researcher, paraprofessional, SLP, peers PMI, visual cues: Prior peer network social skills instruction for peers and participants; direct instruction of target play skills for peers and participants; teacher prompted peers to initiate with participants and to use visual cues or hand over-hand prompts to elicit an appropriate social response from participants during recess; reinforcement card and token economy for all children for talking, sharing, and playing appropriately	Social communication skills: Communicative acts	Tau-U = .98 [.67, 1] NAP = .99 [.68, 1]
McCurdy and Cole (2014)	MBD across participants	3 males; 8, 7, and 11 years High FXN Ethnicity NR	Classroom 5 × per week for up to 3 weeks, *M* = 10 sessions per participant	Researcher, paraprofessional, peers PMI: Peers were trained through modeling and role-play to identify off-task behavior, prompt on-task behavior, and offer participants nonverbal positive feedback; peers prompted on-task behavior and provided performance feedback and encouragement to participants	Classroom behavior: On task behavior	Tau-U = .96 [.62, 1] NAP = .98 [.63, 1]

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Table 1 (*continued*)

Reference	Design	Participant characteristics	Setting and dosage	Agent, intervention type, and procedures	Skill domain and target behaviors	ES[a] and 95% CI
Morrison, Sainato, Benchaaban, and Endo (2002)	MBD across participants	2 males, 2 females; 3, 4, 5 years Moderate FXN Ethnicity NR	Classroom 45 min, 5 × per week, *M* = 37 sessions per participant	Researcher Correspondence training, activity schedules: Participants were trained to use a visual schedule through verbal prompting; participants taught to use play materials through graduated guidance prompting; praise and positive reinforcement given when participants followed activity schedules; correspondence training and verbal feedback used to teach participants to play with what was indicated on the visual schedule	Play skills: On task play, play correspondence	Tau-U = .91 [.60, 1] NAP = .95 [.70, 1]
Owen-DeSchryver, Carr, Cale, and Blakely-Smith (2008)	MBD across participants	3 males; 7 and 10 years High, moderate FXN Ethnicity NR	Playground, cafeteria 25–35 min for up to 14 weeks, *M* = 11 sessions per participant	Researcher, peers PMI: Peers received a rationale for befriending participants and trained in strategies to initiate interaction and engage in play with participants; peers created visual supports of strategies to use during interaction with participants	Social communication skills: Initiations, responses	Tau-U = .65 [.39, .90] NAP = .83 [.58, 1]
Rosenbloom, Mason, Wills, and Mason (2016)	Reversal	1 male; 9 years High FXN African American	Classroom 2–3 × per week, 11 sessions	Researcher Self-monitoring: Before intervention, participant taught to use hand-held tablet device, identify examples and nonexamples of on-task behavior, ask clarifying questions about device, and verbal prompts were given to use device; verbal prompts and adult support to use device withdrawn in intervention	Classroom behavior: On task behavior Challenging behavior: Disruptive behavior	Tau-U = .89 [.43, 1] NAP = .98 [.52, 1] Tau-U = .98 [.52, 1] NAP = .99 [.52, 1]
Sansosti and Powell-Smith (2008)	MBD across participants	3 males; 6, 8, and 9 years High FXN Ethnicity NR	Playground 2 × per week, *M* = 13 sessions per participant	Teacher Social Stories, video modeling: Participants viewed computer implemented Social Stories with peer video models demonstrating target behaviors; adults prompted participants to engage in the target behavior and to peers to engage in the activity requested by the participant	Social communication skills: Joining in, maintaining conversation	Tau-U = .69 [.48, .90] NAP = .87 [.66, 1]
Strain, Wilson, Wilson, and Dunlap (2011)	MBD across participants	2 males, 1 female: 5, 8, and 9 years Moderate, high FXN European American and Hispanic	Classroom 15 min, *M* = 4 sessions per participant	Teacher Function-based intervention and Prevent-Teach-Reinforce (PTR): PTR procedure consisting of FBA, direct instruction of replacement skills, self-monitoring, visual cues, and verbal or tangible reinforcement of replacement behaviors	Challenging behavior: Aggression, elopement, disruptive behavior Classroom behavior: task engagement	Tau-U = .97 [.61, 1] NAP = .98 [.62, 1] Tau-U = 1 [.62, 1] NAP = 1 [.62, 1]

Note. ES = effect size; CI = confidence interval; MBD = multiple baseline design; FXN = functioning level; NR = not reported; PMI = peer mediated intervention; RRB = restricted and repetitive behavior; FBA = functional behavior assessment; PECS = picture exchange communication system; TIP = teaching interaction procedure.

[a] Effect sizes for Tau-U are considered large if above .70, moderate if ranging from .50 to .69, and small if below .50; NAP effect sizes are considered large if above .93, moderate if ranging from .66 to .92, and small if below .65. Cohen's *d* effect sizes of .80, .50, and .20 are considered large, moderate, and small, respectively. Pearson's *r* effect sizes of .50, .30, and .10 are considered large, moderate, and small, respectively.

methods used during treatment sessions (Reichow & Volkmar, 2010). Specific target behaviors were reported and categorized according to skill domain. Domains consisted of social communication skills (i.e., verbal and nonverbal behaviors used to interact and convey meaning with others; Watkins, Kuhn, Ledbetter-Cho, Gevarter, & O'Reilly, 2017), restricted and repetitive behavior (i.e., restricted, repetitive patterns of behavior, interests, or activities; Leekam, Prior, & Uljarevic, 2011), play skills (i.e., behaviors involved in functional or symbolic play; Lang et al., 2009), challenging behavior (i.e., disruptive behaviors such as tantrums, aggression, and self-injury; Machalicek, O'Reilly, Beretvas, Sigafoos, & Lancioni, 2007), academic skills (i.e., skills needed for proficiency in content areas such as language arts, math, science, or social studies; Keen, Webster, & Ridley, 2016), and classroom behavior (i.e., adaptive behaviors needed for regular and successful classroom participation such as hand-raising, sitting at desk, or attending to a teacher or task; Mouga, Almeida, Café, Duque, & Oliveira, 2015).

Finally, the presence or absence of numerous social validity indicators for each study was coded. Studies demonstrated the social validity of the intervention by indicating the following: socially important dependent variable, time and cost-effective intervention, clinically significant behavior change, normative comparisons between individuals with and without disabilities, consumer satisfaction with the results, independent variable manipulated by people typically in contact with the participant, and/or intervention provided in natural contexts (Reichow, Volkmar, & Cicchetti, 2008). The results of the social validity analysis are provided in Table 2.

Intervention Effectiveness

For group design studies, we reported Cohen's *d* and Pearson correlation coefficients to determine intervention effectiveness. Cohen's *d* effect sizes of .80, .50, and .20 are considered large, moderate, and small, respectively (Cohen, 1988). Pearson's *r* effect sizes of .50, .30, and .10 are considered large, moderate, and small, respectively (Cohen, 1992). Two nonparametric measures suitable for SCD studies, Tau-U and nonoverlap of all pairs (NAP), were calculated to provide statistical analysis of intervention results in each study. As there is no single agreed upon effect size estimate suitable for SCD research, we followed recommendations to provide multiple measures along with visual analysis of the data (Maggin & Odom, 2014; Manolov, Gast, Perdices, & Evans, 2014; Whalon, Conroy, Martinez, & Werch, 2015). Tau-U measures the amount of overlap between two phases and is well suited to small data sets typical to SCD research (Parker, Vannest, Davis, & Sauber, 2011). As Tau-U follows the "S" sampling distribution, *p* values and confidence intervals are available. In addition, Tau-U controls for positive data trend (monotonic trend) in the baseline phase (Parker et al., 2011). Effect sizes for Tau-U are considered large if above .70, moderate if ranging from .50 to .69, and small if below .50 (King et al., 2016). NAP measures nonoverlap or dominance across phases and is appropriate for all data types and distributions (Parker et al., 2011). NAP is compatible with visual analysis, relates closely to R2, and produces narrow confidence intervals (CIs), indicating high score precision (Parker et al., 2011). NAP effect sizes are considered large if above .93, moderate if ranging from .66 to .92, and small if below .65.

Table 2
Social Validity Indicators

Study	Socially important DV	Clinically significant behavior change	Occurred in natural context	Delivered by those in typical contact with participant	Consumer satisfaction with results	Time and cost effective intervention	Comparisons with TD peers
Blair et al. (2007)	Yes	Yes	Yes	Yes	Teachers	Yes	Yes
Bock (2007)	Yes	Yes	Yes	Yes	Teachers, participant	Yes	Yes
Chan and O'Reilly (2008)	Yes	Yes	Yes	No	Parents, teachers, teaching assistants	Yes	No
Cihak, Fahrenkrog, et al. (2010)	Yes	Yes	Yes	Yes	Teachers, participants	Yes	No
Cihak, Wright, et al. (2010)	Yes	Yes	Yes	Yes	Teachers, participants	Yes	No
Clemons et al. (2016)	Yes	Yes	Yes	No	Teachers, participants	NR	No
Ganz and Flores (2008)	Yes	Yes	Yes	Partially	NR	NR	No
Gardner et al. (2014)	Yes	Yes	Yes	Yes	Parents, participants, school staff, peers	Yes	Yes
Gena (2006)	Yes	Yes	Yes	Yes	Parents, teachers	NR	Yes
Haley et al. (2010)	Yes	Yes	Yes	Yes	Teachers, paraprofessionals	Yes	No
Hochman et al. (2015)	Yes	Yes	Yes	Yes	Parents, participants, staff, peers	Yes	Yes
Hughes et al. (2011)	Yes	Yes	Yes	Partially	Peers	NR	Yes
Hughes et al. (2013)	Yes	Yes	Yes	Partially	Participants, peers, teachers	Yes	Yes
Jung et al. (2008)	Yes	Yes	Yes	Partially	Parents, teachers	Mixed results	Yes
Kasari et al. (2012)	Yes	Yes	Yes	Partially	NR	NR	No
Kasari et al. (2016)	Yes	Yes	Yes	Partially	NR	NR	No
Kern and Aldridge (2006)	Yes	Yes	Yes	Yes	NR	No	Yes
Koegel et al. (2012)	Yes	Yes	Yes	No	NR	Yes	Yes
Koegel et al. (2013)	Yes	Yes	Yes	No	Participants, peers	Yes	Yes
Kretzmann et al. (2015)	Yes	Yes	Yes	Yes	NR	NR	No
Loftin et al. (2008)	Yes	Yes	Yes	Partially	Parents, teachers, peers	Yes	No
Mason et al. (2014)	Yes	Yes	Yes	Partially	Speech language pathologist, paraprofessional	Mixed results	No
McCurdy and Cole (2014)	Yes	Yes	Yes	Partially	Teachers, participants, peers	Yes	Yes
Morrison et al. (2002)	Yes	Yes	Yes	No	Parents, teachers	Yes	No
Owen-DeSchryver et al. (2008)	Yes	Yes	Yes	Partially	NR	NR	Yes
Rosenbloom et al. (2016)	Yes	Yes	Yes	No	Teacher, participant	Yes	No
Sansosti and Powell-Smith (2008)	Yes	Yes	Yes	Yes	Teachers	Yes	Yes
Strain et al. (2011)	Yes	Yes	Yes	Yes	School staff	Yes	No
% per indicator	100%	100%	100%	43%[a]	75%	61%	50%

Note. DV = dependent variable; TD = typically developing; NR = not reported.
[a] Percentage of interventions solely delivered by those in typical contact with participants (e.g., teachers, paraprofessionals, and peers).

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To enable effect size calculation for SCD studies, graphed data from each study were extracted and the raw data were entered into a spreadsheet and organized by phase. Effect sizes for individual studies were calculated by contrasting data from adjacent AB phases and weighting the results (Chen, Hyppa-Martin, Reichle, & Symons, 2016). Because generalization and maintenance of skills are important indicators of the overall effectiveness of an intervention (Bellini, Peters, Benner, & Hopf, 2007; Watkins et al., 2015), these data were also compared with adjacent phases and included in effect size calculations. Data were entered into a Web-based tool (www.singlecaseresearch.org) to calculate an effect size (Vannest, Parker, & Gonen, 2011). For Tau-U calculations, baseline trends were first examined and adjusted for levels of 0.4 or above in the direction associated with treatment before phase comparisons were made (King et al., 2016; Parker et al., 2011). These effect sizes were meta-analyzed via the MedCalc (v. 12.75; MedCalc Statistical Software, 2013) software and weighted according to number of participants per study.

In addition to calculating effect sizes for individual studies, average effect sizes across different study features (i.e., participant age, gender, race, and functioning level; intervention type; intervention agent; skill domain; treatment intensity; and social validity indicators) were examined to identify potential moderating influences. Because the limited number of group design studies did not allow for further inferential analysis, we included only SCD studies in these calculations. Statistical significance of differences between groups was determined by performing the Mann–Whitney *U* test using the Statistical Package for Social Sciences (SPSS; Mann & Whitney, 1947). The Mann–Whitney test, essentially a nonparametric version of a *t* test, is appropriate for data that are not normally distributed, like the effect sizes of the studies included in this meta-analysis (McKnight & Najab, 2010). We have also noted which comparisons remained statistically significant after applying a Bonferroni correction to reduce the likelihood of committing a Type I error, whereby the α level of .05 was decreased by dividing this value by the number of comparisons made. Table 3 reports average Tau-U and NAP scores across different variables and notes when comparisons are statistically significant.

Reliability

Initially, the first author used data extraction methods to code the studies and develop an initial summary of each study. Ten studies (36%) were randomly selected to measure the accuracy of these summaries through independent verification by a coauthor using a checklist that included questions regarding accuracy in summaries of the following: (a) study design, (b) participant characteristics, (c) intervention setting, (d) intervention dosage, (e) intervention agent, (f) intervention type, (g) intervention procedures, (h) skill domain and target behavior, (i) intervention effect sizes, and (j) social validity. Interrater agreement was calculated across 100 items on which there could be agreement or disagreement (i.e., 10 studies with 10 questions per study). Initial agreement was obtained on 94 items (94%), and discrepancies were discussed between authors until 100% agreement was obtained.

Results

Twenty-eight focused intervention studies met the inclusion criteria for this meta-analysis. Table 1 provides descriptive summaries and results of each study, Table 2 provides social validity findings, and Table 3 provides effect sizes for study variables.

Table 3
Effect Sizes for Study Variables

Study variables	*n*	Mean Tau U (*SD*)	Mean NAP (*SD*)
Participant functioning level			
Lower	13	.88 (.15)	.94 (.07)
Medium	21	.87 (.15)	.93 (.07)
Higher	46	.88 (.14)	.95 (.07)
Participant age			
Pre-K/Kindergarten	27	.89 (.15)	.95 (.07)
Elementary	29	.90 (.14)	.95 (.06)
Secondary school	29	.86 (.14)	.94 (.07)
Participant gender			
Male	74	.88 (.14)	.94 (.07)
Female	11	.90 (.13)	.95 (.06)
Participant race			
White	17	.87 (.14)	.94 (.07)
Latino	9	.79 (.18)	.91 (.09)
African American**	5	.98 (.01)	.99 (.00)
Asian**	3	.84 (.18)	.93 (.09)
Biracial**	2	.64 (.05)	.84 (.00)
Intervention type[a]			
Peer-mediated	31	.87 (.17)*	.93 (.08)*
Visual supports	31	.90 (.15)*	.95 (.07)*
Peer network	16	.79 (.12)*	.91 (.06)*
Self-monitoring	12	.94 (.07)*	.97 (.03)*
Function-based	10	.98 (.03)*	.99 (.01)*
Intervention agent[b]			
Teacher	27	.95 (.10)*	.97 (.04)*
Researcher	21	.87 (.11)*	.94 (.05)*
Researcher and peer	21	.83 (.18)*	.91 (.09)*
Teacher and peer	10	.81 (.16)*	.91 (.08)*
Teacher, researcher, and peer**	6	.97 (.04)	.98 (.02)
Skill class			
Social communication	51	.83 (.16)*	.92 (.08)*
Classroom behavior	19	.97 (.05)*	.98 (.02)*
Challenging behavior**	6	.98 (.02)	.99 (.01)
Play**	5	.92 (.06)	.96 (.03)
Restrictive/repetitive behavior**	4	.99 (.01)	.99 (.00)
Intensity			
1–2 h	10	.97 (.02)	.98 (.02)
3–4 h	20	.83 (.18)	.92 (.08)
5–6 h	16	.86 (.15)	.94 (.07)
6 or more hours	15	.84 (.12)	.93 (.05)
Social validity			
6–7 indicators	46	.89 (.13)	.95 (.06)
4–5 indicators	36	.90 (.12)	.95 (.05)
3 or less indicators**	3	.60 (.25)	.81 (.12)

[a] NAP scores for intervention type: FBA > all other groups; visual supports and self-monitoring > peer networks; Tau U scores for intervention type: peer networks < all other intervention groups; FBA > peer mediated. [b] NAP and Tau U scores for intervention agent: teachers > all other groups with enough cases for analysis.
* $p < .05$ level. ** Not enough cases for statistical analysis.

Participant Characteristics

A total of 293 students with ASD were included in this meta-analysis. Participants were predominantly male ($n = 243$; 83%), with relatively fewer female participants represented ($n = 50$; 17%). The vast majority of participants were elementary school age ($n = 243$; 83%). Twenty-seven participants (9%) were secondary school age and 23 participants (8%) were preschool age.

Roughly half of the studies (n = 13; 46%) did not specify the race, ethnicity, or cultural background of 38 participants included in this meta-analysis. Of the 15 studies (54%) that reported the race, ethnicity, or cultural background of the other 255 participants, most were described as White, Caucasian, or European American (n = 126; 49%), followed by participants described as Hispanic or Latino (n = 44; 17%), Asian or Asian American (n = 40; 15%), Black or African American (n = 20; 8%), and biracial (n = 2; 1%). Sixteen participants (6%) described their ethnicity as "other" and 10 participants (4%) preferred not to disclose this information.

Two studies did not provide information that allowed for an estimation of functioning level for 25 participants (8.5%). Ninety-six participants (33%) were described as higher functioning, with well-developed verbal communication skills or an IQ above 85. Of the higher functioning participants, 88 were male and 8 were female. Twenty-two participants (7.5%) were described as moderate functioning, with basic verbal communication skills or an IQ of 55–85. Of the moderate functioning participants, 18 were male and 4 were female. Two studies reported that participants involved in the intervention were either high or moderate functioning (n = 137; 47%) but did not specify the number of participants within each category. Thirteen participants (4%) were described as lower functioning, with limited or no verbal language skills or an IQ less than 55. Of the lower functioning participants, 9 were male and 4 were female.

Setting and Intervention Dosage

Interventions took place in a classroom (n = 18), a playground or schoolyard (n = 12), cafeteria (n = 5), and other school locations such as a playroom or hallway (n = 2). Six studies used a combination of these settings (e.g., Hughes et al., 2011; Koegel, Kim, Koegel, & Schwartzman, 2013). The duration of sessions was not reported in seven studies, precluding the calculation of treatment intensity. Of the 20 studies reporting the duration of the intervention sessions, treatment intensity varied considerably. Most studies (n = 7) implemented a total intervention time of 3 to 4 h per participant, followed by 5 to 6 h (n = 5), 6 or more hours (n = 5), and 1 to 2 h (n = 3). Duration of intervention sessions ranged from 5 to 90 min and occurred from once a week to five times a week.

Intervention Agent, Type, and Procedures

Teachers or other school staff such as paraprofessionals served as the sole intervention agents in most studies (n = 9; 32%), followed by researchers and peers (n = 8; 28.5%), researchers (n = 6; 21.5%), teachers and peers (n = 3; 11%), and a combination of researchers, teachers/school staff, and peers (n = 2; 7%).

Intervention type varied across studies. Ten studies (36%) used some form of visual support strategy or strategies to cue, prompt, or reinforce behaviors (Quill, 1997; Reichow & Volkmar, 2010). Specifically, these included visual cues (n = 4), social scripts (n = 3), Social Stories (n = 2), video modeling (n = 2), and communication books (n = 1). Eleven studies (39%) used PMI, in which peers were directly trained to initiate, reinforce, and maintain interaction with participants with ASD (Chan et al., 2009; Watkins et al., 2015). Four studies (14%) used self-monitoring strategies, in which participants were taught to observe and record specific behaviors (Ganz & Sigafoos, 2005). Four studies (14%) used peer networks, in which a cohesive group of peers led by an adult facilitator interacted regularly with a participant with ASD within the context of a shared social activity (DiSalvo & Oswald, 2002). Three studies (11%) developed individualized interventions based on the results of a functional behavior assessment (FBA), which assesses the relationship between the environment and the student's behavior (O'Neill et al., 1997). Two studies (7%) implemented social skills groups in which two or more students with ASD received group instruction of skills (Reichow et al., 2008; Williams White, Keonig, & Scahill, 2007). Other strategies such as initiations training, high probability request sequences, music therapy, and unspecified behavioral strategies were implemented in one study each.

Multiple intervention types were frequently combined to create unique treatment packages. For example, PMI was often used in combination with strategies such as script training (Ganz & Flores, 2008), communication books (Hughes et al., 2011), high probability request sequences (Jung, Sainato, & Davis, 2008), music therapy (Kern & Aldridge, 2006), and self-monitoring (Loftin, Odom, & Lantz, 2008). Similarly, visual support strategies were often implemented alongside intervention strategies such as role-play (Chan & O'Reilly, 2008), self-monitoring (Cihak, Wright, & Ayres, 2010), PMI (Mason et al., 2014), and the teaching interaction procedure (Hughes et al., 2011). Regardless of the type of intervention used, each study incorporated procedures based on the principles of applied behavior analysis, namely prompting (i.e., antecedent procedure to increase the likelihood that a specific behavior will occur; Wong et al., 2015), modeling (i.e., adult or peer demonstrations of a target behavior that should result in imitation of the behavior by the participant; Wong et al., 2015), and/or reinforcement (i.e., a consequence that increases the likelihood that a behavior will be performed again; Wong et al., 2015). These strategies were not only incorporated into intervention procedures across all studies but were also frequently used by researchers to train teacher or peer interventionists to deliver treatment to participants with ASD (e.g., Jung et al., 2008; Kern & Aldridge, 2006; McCurdy & Cole, 2014; Strain, Wilson, Wilson, & Dunlap, 2011).

Skill Domain and Target Behaviors

The vast majority of studies (n = 22; 79%) targeted improvements in social communication skills. These studies measured discrete social communication behaviors such as interaction with peers (n = 9), initiations or comments (n = 8), social or peer engagement (n = 4), responses to initiations (n = 4), communicative acts (n = 3), and maintaining conversation (n = 1). Eleven studies (39%) measured multiple social communication variables within the context of a single experiment (e.g., initiations and responses or initiations and social engagement). Two RCTs measured social improvement through standardized measures such as the Social Network Survey, Teacher Perception of Social Skills, Friendship Survey, and Student Teacher Relationship Scale.

Nine studies (32%) targeted improvement in adaptive behaviors needed for success in the classroom. These included task engagement or on-task behavior (n = 6), following directions (n = 1), cooperative learning participation (n = 1), hand-raising (n = 1),

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and transitioning between activities ($n = 1$). One study (4%) measured multiple adaptive classroom behaviors within the context of a single experiment (e.g., both task engagement and following teacher directions).

Five studies (18%) targeted a reduction in challenging behavior. These included aggression ($n = 2$), inappropriate vocalizations ($n = 2$), disruptive behaviors ($n = 2$), out of seat behaviors ($n = 1$), inappropriate interactions ($n = 1$), conflict ($n = 1$), and elopement ($n = 1$). Three studies (12%) measured multiple challenging behaviors within the context of a single experiment (e.g., aggression, elopement, and disruptive behavior).

Two studies (7%) targeted a reduction in restricted and repetitive behaviors, which included vocal stereotypy ($n = 1$) and motor stereotypy ($n = 1$). Two studies (8%) targeted improving play skills, including game playing ($n = 1$) and play correspondence ($n = 1$). Several studies examined social communication skills within the context of play ($n = 9$) or leisure activities ($n = 4$) but did not collect or report data on play behaviors other than as relating to interactions or engagement with peers.

Seven studies (25%) targeted variables across multiple skill domains (e.g., both social communication skills and challenging behavior or both social communication skills and classroom behavior). Zero studies included academic outcomes that measured success in specific content areas.

Intervention Effectiveness

Cohen's *d* and Pearson's *r* for each of the three RCTs are reported in Table 1. When possible, CI_{95} was calculated for these studies. Effect sizes for both Cohen's *d* and Pearson's *r* indicated small to large treatment effects. One study indicated small effects, one study indicated small to moderate effects, and one study indicated large effects.

NAP and Tau-U effect sizes for each of the 25 SCD studies are reported in Table 1. A weighted mean effect size with CI_{95} was reported for outcomes within each skill domain. If studies reported outcomes across multiple skill domains (e.g., Bock, 2007; Chan & O'Reilly, 2008; Loftin et al., 2008), an effect size was calculated for each domain. Thus, a total of 34 Tau-U and NAP effects were calculated. Effect sizes for both Tau-U and NAP ranged from moderate to large. Twenty-nine Tau-U outcomes indicated large effects (85%) and five indicated moderate effects (15%). This was highly comparable with NAP scores, with 28 outcomes indicating large effects (82%) and 6 indicating moderate effects (18%). The reported effect sizes aligned closely with visual analysis interpretation of results, which were established among studies during the initial screening for inclusion.

Because of the limited number of group design studies ($n = 3$) further inferential analysis was not possible for the RCTs. Table 3 reports SCD effect sizes when averaged across specific study variables and analyzed for statistical significance. After applying a Bonferroni correction to reduce the likelihood of Type I errors, several variables remained statistically significant. When examining the impact of intervention type, analysis of NAP scores indicated that treatment packages utilizing FBA procedures produced significantly better outcomes than all other intervention types. Additionally, visual supports and self-monitoring interventions resulted in significantly higher effects than peer networks. Analysis of Tau-U scores revealed that FBA interventions resulted in significantly better outcomes than PMI and that all interventions produced significantly higher effects than those involving peer networks. Regarding the intervention agent, teachers who delivered the intervention by themselves produced results with significantly higher effects than interventions delivered by any other agent(s). Studies targeting classroom behaviors resulted in significantly larger effects than those targeting social communication.

Given that NAP values are a percentage ranging from 0 to 100%, we meta-analyzed values of Tau-U ranged from 0.65 to 1.00 with a mean effect size of 0.91 (95% CI [0.97, 0.99]). We weighted SCDs according to the number of SCDs per study so that a study with more than one effect size would not skew results. Results indicate significant heterogeneity among studies, $Q(33) = 544.69$, $p < .001$, $I^2 = 93.94\%$. This high value of I^2 indicates that approximately 94% of the total variation is because of variation between single-case designs. This high degree of variation among SCDs is not unexpected given the variety of operational definitions of the related outcome variables of social communication, challenging behavior, and classroom behavior. Given this variability among studies, the mean Tau-U value under a random effects model was reported indicating a value of 0.991, $p < .001$. To address the issue of potential publication bias, we calculated a fail-safe *n* value utilizing Orwin's (1983) formula, which revealed a value of 18 based upon a Tau-U criterion value of 0.65. The lowest value of Tau-U among the 34 SCDs was considered to be the threshold for clinically or practically significant for the purposes of the current study. This value indicates that approximately 18 null SCDs would have to exist to negate the results of the meta-analysis below the Tau-U criterion value. We next calculated the conditional probability of these additional 18 null SCDs existing using Bayes's theorem. We found that there was approximately a 12% probability of an additional 18 null SCDs existing. Given this relatively low probability of an additional 18 null SCDs existing, we consider the potential publication bias to be relatively low.

We next aggregated SCDs according to the outcome of social communication. In testing the heterogeneity among these SCDs, the degree of heterogeneity was still significant, $Q(18) = 84.57$, $p < .001$, yet lower with an I^2 value of 79.90%. This value still indicates a high degree of heterogeneity among SCDs yet less so, which is expected when delimiting our analyses to just the outcome of social communication. The mean Tau-U value was 0.94 (95% CI [0.87, 0.97]). We also aggregated SCDs according to the outcome of behavior because of the limited number of SCDs (i.e., collapsing classroom behavior, challenging behavior, and repetitive behavior). In testing the heterogeneity among these SCDs, the degree of heterogeneity was significant, $Q(15) = 304.65$, $p < .001$ and quite high with an I^2 value of 95.08%. This value indicates a high degree of heterogeneity among SCDs but is not unexpected given that we collapsed related but different behaviors. The mean Tau-U value was 0.99 (95% CI [0.98, 1.00]).

Social Validity Indicators

Table 2 provides social validity indicators for each study. A majority of the included studies ($n = 25$; 89%) demonstrated adequate social validity (i.e., four or more indicators). All studies indicated the social importance of the dependent variable(s), demonstrated a clinically significant behavior change, and occurred in

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the participants' natural context. Most studies (n = 21; 75%) reported consumer (i.e., teachers, participants, parents, or peers) satisfaction with the results of the intervention, with one study reporting mixed results and four not reporting this information. Seventeen studies (61%) confirmed the time and cost effectiveness of the intervention, with two studies reporting mixed results and eight not reporting this information. Fourteen of the 28 studies (50%) provided comparisons between participants with ASD and their typically developing classmates. Less than half of the studies (n = 12; 43%) were delivered solely by individuals in the participants' natural context (e.g., teacher, paraprofessionals, and peers). However, 10 studies (36%) were delivered both by researchers and individuals in the natural environment.

Discussion

This meta-analysis is the first comprehensive work to quantitatively analyze the effectiveness of focused interventions for students with ASD in inclusive classroom settings using methodologically rigorous research studies. Results suggested that these interventions have generally produced moderate to large effects across skill domains, with the majority of interventions found to be socially valid in this setting. Confidence in the positive outcomes reported is supported by the fact that studies met the rigorous methodological criteria outlined in the WWC, though these findings should be interpreted judiciously given the small number of studies. Most of the included studies (n = 22; 79%) targeted social communication skills, involved predominantly higher or moderate functioning participants (n = 255; 87%), and, among the SCD studies, reported mostly large effects for both Tau-U (n = 29; 85%) and NAP (n = 28; 82%). Variables found to significantly influence treatment effectiveness included intervention type, intervention agent, and the class of skill targeted.

Most intervention types resulted in large effects. Function-based interventions using FBA procedures to inform the development of an individualized treatment plan resulted in significantly better outcomes than all other intervention types. FBA is considered an evidence-based practice according to both WWC (2013) and NPDC (Wong et al., 2015) and should also be considered an evidence-based strategy for students with ASD in inclusive classrooms. Visual supports, self-monitoring, and PMI, which are recognized as evidence-based practices by NPDC, resulted in large effects and should also be considered recommended interventions for this student population. These large effects could be because of providing treatment in the natural setting, which supports the generalization and maintenance of skills, in addition to the incorporation of behavioral strategies such as modeling, prompting, and reinforcement of target skills (Bellini et al., 2007; Ganz & Ayres, 2018). Indeed, these interventions reported positive generalization and maintenance effects, which in part contributed to overall large effects. By contrast, peer networks, when taking into consideration generalization and maintenance data, produced less robust effects overall. These interventions tended to include less explicit instruction and prompting and reinforcement of target behaviors, which may have produced less robust generalization results. Although peer networks resulted in positive outcomes, additional strategies that promote the generalization and maintenance of skills may be needed when using this intervention.

Teachers or other school staff acted as intervention agents in nine studies (32%) and produced outcomes significantly better than any other intervention agent(s). This finding is encouraging and demonstrates that teachers in inclusive classrooms are able to implement effective and socially valid interventions for students with ASD. Given that general education teachers typically do not have training in or knowledge of research-based interventions for students with ASD (Pazey, Gevarter, Hamrick, & Rojeski, 2014; Segall & Campbell, 2012), and as the number of students with ASD included in these settings continues to grow, teacher preparation programs and school administrations should provide the necessary resources and training that will allow general and special education teachers in inclusive settings to successfully address these students' needs.

Age, gender, and functioning level did not influence outcomes, which offers practitioners a variety of intervention approaches for most subsets of the population. Similarly, the intensity or duration of treatment did not impact results. Shorter interventions of 1–2 h produced strong effects, which should be attractive to practitioners working with these students. Higher functioning students were most commonly represented in this work (51%) and may be more likely to be included in inclusive settings (Watkins, O'Reilly, et al., 2017). These students perhaps more often present with a performance deficit (i.e., lack of opportunity for skill use) rather than a skill deficit (i.e., lack of ability). Thus, higher functioning students may not need interventions as time-intensive as those with skills deficits, for whom more exposure may be required to directly teach a certain skill or behavior. It is important to note, however, that these results did not reach statistical significance and are not causal. These findings should be considered cautiously, and more research is needed to examine potential moderators of intervention effectiveness.

In addition to these recommendations for practice, there are several evident directions for future research. That most interventions analyzed in this meta-analysis targeted social communication skills is unsurprising as social communication impairments are present in individuals with ASD regardless of cognitive or language ability and are often cited as top treatment concerns (Pituch et al., 2011). As improved social communication skills may produce positive collateral effects (Ledbetter-Cho, Lang, Watkins, O'Reilly, & Zamora, 2017), it would seem prudent for studies targeting social communication to also assess whether such interventions produce improvement in other areas. For example, Loftin et al. (2008) found that PMI and self-monitoring strategies to improve social interaction resulted in a collateral decrease in the motor stereotypy of participants. Similarly, Blair, Umbreit, Dunlap, and Jung (2007) found that a function-based intervention to improve social communication skills led to collateral decreases in challenging behavior and an increase in appropriate classroom behaviors such as task engagement. Interventions that potentially produce multiple desirable outcomes may increase the impetus for teachers in inclusive classrooms to use these strategies.

That few studies assessed outcomes for restricted and repetitive behavior, play skills, challenging behavior, and none targeted academic skills in inclusive settings is cause for concern. Deficits in play skills and the presence of restricted and repetitive behaviors or interests are both defining features of ASD (APA, 2013), and it seems likely that interventions for these behaviors are needed for students with ASD in inclusive settings. Similarly, challenging

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behavior is prevalent in children with ASD and is often considered an associated feature of the diagnosis (Matson & Nebel-Schwalm, 2007), yet scant research focusing on interventions to treat challenging behavior of students with ASD in inclusive classrooms exists. Challenging behavior can be a barrier to successful inclusion (Crosland & Dunlap, 2012; Dunlap et al., 2010; Emerson et al., 2001), so it would be prudent for future research to more thoroughly assess which strategies are effective in reducing challenging behavior and are feasible to implement in inclusive classrooms. Additionally, many students with ASD experience difficulties with academic functioning, and research has shown that teachers struggle with identifying and implementing academic interventions for this population (Whitby & Mancil, 2009). Thus, there is a great need for research-based interventions that will help improve academic outcomes for students with ASD. Although academic focused interventions for students with ASD exist (see El Zein, Solis, Vaughn, & McCulley, 2014; Gevarter et al., 2016; and King et al., 2016 for reviews of the literature), these studies did not take place in inclusive settings or failed to meet the research methodology standards required to be included in this meta-analysis.

Research must also strive to include participants with more diverse characteristics, especially female participants and those from diverse racial and ethnic backgrounds. The scarcity of female participants reflects a wider concern in the field about the underrepresentation of females with ASD in research and current diagnostic procedures that may fail to identify females with ASD (Kirkovski, Enticott, & Fitzgerald, 2013; Loomes, Hull, & Mandy, 2017; Ratto et al., 2018). With more accurate identification and diagnostic tools on the horizon, it will be necessary to develop interventions that meet the particular needs of female students as they may present with different characteristics compared with males with ASD (Ratto et al., 2018). Researchers should also strive to both report the ethnic and racial backgrounds of participants as well as include more racially and ethnically diverse participants in studies. Forty-six percent of studies did not report the background of participants, which mirrors the widespread lack of race and ethnicity reporting in autism research (Pierce et al., 2014). Of the participants whose racial and ethnic background was reported, the majority were White. Factors such as race and ethnicity may serve as mitigating variables and should be considered when designing interventions to meet the needs of diverse participants (Mandell, Listerud, Levy, & Pinto-Martin, 2002; Pierce et al., 2014).

The majority of studies (89%) demonstrated a sufficient level of social validity (i.e., four or more indicators). Rather than only assessing social validity post intervention (e.g., practitioner satisfaction with intervention results), researchers should consider directly programming for multiple measures of social validity when developing interventions for students with ASD in inclusive classrooms (e.g., solicit teacher input when selecting skills to teach or intervention components to include and develop). Furthermore, only 43% of the interventions were implemented solely by those in the participants' natural environment. Researchers should develop and report processes for training teachers and staff to implement interventions in replicable detail. Interventions that practitioners in these settings consider time and cost-effective, target outcomes appropriate to the inclusive classroom setting, and provide measures of comparison with typically developing students may increase teacher "buy in" and their willingness to implement intervention procedures with fidelity. That is, teachers may be more likely to adopt interventions that contain these components and carry them out as designed (Kennedy, 2002; Lang & Page, 2011; Pazey et al., 2014).

The findings of this meta-analysis provide recommendations for both practice and research, yet results should be cautiously considered because of the following limitations. The interpretation of results must be considered in relation to the established confidence intervals for each outcome. For example, a study reporting a moderate effect with a relatively narrow CI may produce results similar to a study reporting a large effect with a wider confidence interval. In addition, although it is necessary to include only those studies with sufficient rigor and evidence of effectiveness within a best-evidence synthesis, reliance on the WWC standards for this purpose is a potential limitation. This framework appears to be used predominantly in the United States as this is where these standards were developed. An artifact of using the WWC standards is that most studies (93%) derived from the United States. However, other agreed upon standards as to academic rigor do not appear to be readily available. Another limitation that emerges from using the WWC standards is the exclusion of less rigorous or flawed studies. Such studies can still provide information while reducing the influence of publication bias, but their incorporation into any synthesis of the literature is often problematic. Furthermore, the majority of studies included in this meta-analysis are SCDs. Though SCD research makes up the majority of the literature base in the field of special education (Maggin & Chafouleas, 2013; Odom et al., 2005; Odom & Strain, 2002), evidence from SCD research is generally not as strong as that of RCTs, and we are unable to conclude that the moderators examined are causal; thus, caution in interpretation of these results must be exercised. Finally, although the higher level of heterogeneity that accompanies SCD research is not unexpected, this limits the generalizability of results.

Despite these limitations, this work provides guidance to both researchers and practitioners. Overall, the results of this meta-analysis indicate that interventions for students with ASD have produced moderate to large effects and have demonstrated a sufficient level of social validity within inclusive school settings. Function-based interventions, visual supports, self-monitoring, and PMI all produced large effects and should be considered recommended strategies for use with students with ASD in inclusive classrooms. The training of teachers to deliver these interventions with fidelity should be prioritized, and there is a need to develop detailed training procedures to teach these intervention strategies to practitioners. Interventions that target domains beyond social communication skills are also needed, as is research that more closely examines potential collateral effects of these interventions. The inclusion of participants with ASD with a broader range of characteristics, notably female students and students with greater ethnic and racial diversity, is also needed. Finally, additional large-scale and rigorously designed RCTs assessing outcomes of focused interventions for students with ASD in inclusive school settings are needed to strengthen the findings reported in this meta-analysis and would seem a logical next step in the field.

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